ASSIGNMENT – Question 3

Comparison of Datasets Based on Standard Deviation

Introduction

In statistics, two fundamental measures used to describe a dataset are the **mean (μ)** and the **standard deviation (σ)**. The mean represents the average value, while the standard deviation indicates how much the data values are spread around that average.

When multiple datasets have the same mean, the standard deviation becomes the key factor in determining which dataset is more stable and reliable. A smaller standard deviation shows that values are closely grouped around the mean, while a larger standard deviation indicates greater variability.

In this problem, we are given:

Mean (μ) = 55
Standard deviations:
$\sigma_a = 4$
$\sigma_\beta = 10$
$\sigma c = 15$

The objective is to identify which dataset is better.

Understanding the Given Values

All three datasets have the same mean (55).
This means the average value of each dataset is identical.

Since the mean is constant, the comparison depends entirely on the standard deviation.

Standard deviation tells us:

- How consistent the data is
- How far values deviate from the average
- How predictable the dataset is

Smaller σ → Less variation → More consistency
Larger σ → More variation → Less consistency

Analysis of Each Case

Case A: $\sigma_a = 4$

This is a small standard deviation.
It means the data values are very close to the mean (55).

Using $\mu \pm \sigma$:

$55 \pm 4 \rightarrow$ Range: 51 to 59

Most values fall within a narrow interval.
This indicates high stability and reliability.

Case B: $\sigma_\beta = 10$

This represents moderate variation.

$55 \pm 10 \rightarrow$ Range: 45 to 65

Values are more spread out compared to Case A.
Consistency is lower than σ_a but better than σc.

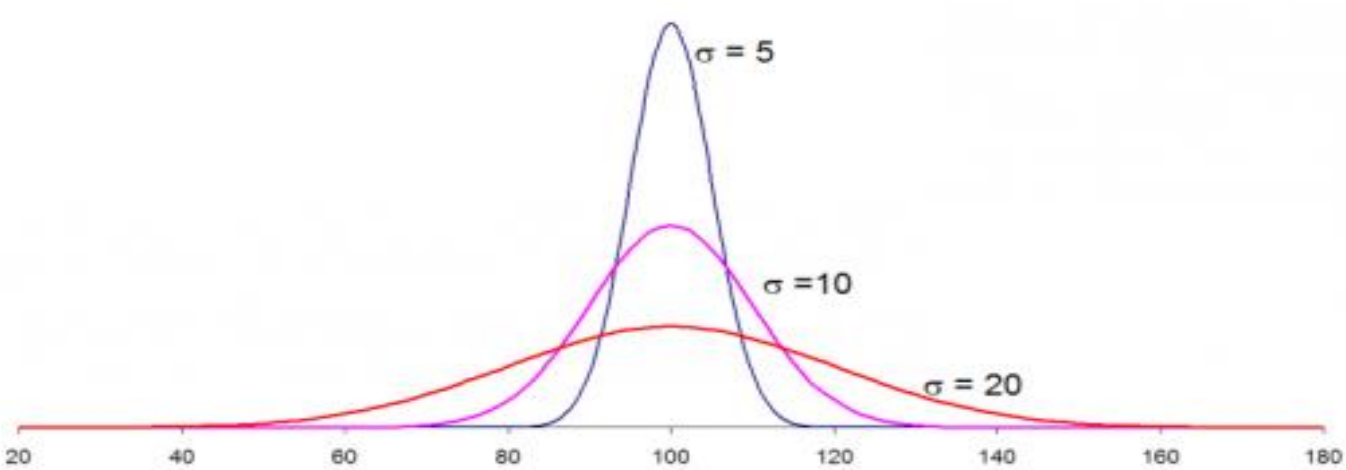
Case C: $\sigma c = 15$

This is a large standard deviation.

$55 \pm 15 \rightarrow$ Range: 40 to 70

Values are widely dispersed around the mean.
This dataset shows higher variability and lower consistency.

Graphical Comparison



In the graph:

- $\sigma_a = 4$ produces a tall and narrow curve (high concentration).
- $\sigma_\beta = 10$ produces a moderately wide curve.
- $\sigma c = 15$ produces a wide and flat curve (high spread).

The narrower the curve, the more consistent the dataset.

Comparison Table

Dataset A
Standard Deviation: 4
Interpretation: Very high consistency

Dataset B
Standard Deviation: 10
Interpretation: Moderate consistency

Dataset C
Standard Deviation: 15
Interpretation: Low consistency

Which Dataset is Better?

Since all datasets share the same mean (55), the dataset with the **smallest standard deviation** is considered better.

Therefore,

$\sigma_a = 4$ is the best dataset

Reason:

- Values are closest to the mean
- Less variation
- More reliable
- Higher predictability
- Greater stability

Importance of Smaller Standard Deviation

A smaller standard deviation is preferred in many real-world situations such as:

- Quality control in manufacturing
- Performance measurement

- Financial risk assessment
- Scientific experiments
- Educational performance analysis

Consistency is often more important than just having a high average.

Conclusion

When comparing datasets with the same mean, standard deviation becomes the deciding factor. A smaller standard deviation indicates that data values are tightly clustered around the mean, resulting in greater consistency and reliability.

In this case, among $\sigma_a = 4$, $\sigma_\beta = 10$, and $\sigma c = 15$, the dataset with $\sigma_a = 4$ is the best because it shows the least variation and the highest stability.

Thus, the dataset with standard deviation 4 is the most dependable and preferable distribution.